

ABEKA SCHOLAR ACADEMY

High School Science Curriculum

Official Curriculum & Study Material

Chapter 1

Welcome to our advanced study module on Quantum mechanics. We have a massive amount of material to cover. Let us begin. Quantum mechanics is the fundamental physical theory that describes the behavior of matter and of light; its unusual characteristics typically occur at and below the scale of atoms. It is the foundation of all quantum physics, which includes quantum chemistry, quantum biology, quantum field theory, quantum technology, and quantum information science. Quantum mechanics can describe many systems that classical physics cannot. Classical physics can describe many aspects of nature at an ordinary (macroscopic and (optical) microscopic) scale, however is insufficient for describing them at very small submicroscopic (atomic and subatomic) scales.

Classical mechanics can be derived from quantum mechanics as an approximation that is valid at ordinary scales. Quantum systems have bound states that are quantized to discrete values of energy, momentum, angular momentum, and other quantities, in contrast to classical systems where these quantities can be measured continuously. Measurements of quantum systems show characteristics of both particles and waves (wave–particle duality), and there are limits to how accurately the value of a physical quantity can be predicted prior to its measurement, given a complete set of initial conditions (the uncertainty principle). Quantum mechanics arose gradually from theories to explain observations that could not be reconciled with classical physics, such as Max Planck's solution in 1900 to the black-body radiation problem, and the correspondence between energy and frequency in Albert Einstein's 1905 paper, which explained the photoelectric effect.

These early attempts to understand microscopic phenomena, now known as the "old quantum theory", led to the full development of quantum mechanics in the mid-1920s by Niels Bohr, Erwin Schrödinger, Werner Heisenberg, Max Born, Paul Dirac and others. The modern theory is formulated in various specially developed mathematical formalisms. In one of them, a mathematical entity called the wave function provides information, in the form of probability amplitudes, about what measurements of a particle's energy, momentum, and other physical properties may yield. == Overview and fundamental concepts == Quantum mechanics allows the calculation of properties and behaviour of physical systems. It is typically applied to microscopic systems: molecules, atoms and subatomic particles. It has been demonstrated to hold for complex molecules with thousands of atoms, but its application to human beings raises philosophical problems, such as Wigner's friend, and its application to the universe as a whole remains speculative.

Predictions of quantum mechanics have been verified experimentally to an extremely high degree of accuracy. For example, the refinement of quantum mechanics for the interaction of light and matter, known as quantum electrodynamics (QED), has been shown to agree with experiment to within 1 part in 10¹² when predicting the magnetic properties of an electron. A fundamental feature of the theory is that it usually cannot predict with certainty what will happen, but only gives probabilities. Mathematically, a probability is found by taking the square of the absolute value of a complex number, known as a probability amplitude. This is known as the Born rule, named after physicist Max Born. For example, a quantum particle like an electron can be described by a wave function, which associates to each point in space a probability amplitude. Applying the Born rule to these amplitudes gives a probability density function for the position that the electron will be found to have when an experiment is performed to measure it.

This is the best the theory can do; it cannot say for certain where the electron will be found. The Schrödinger equation relates the collection of probability amplitudes that pertain to one moment of time to the collection of probability amplitudes that pertain to another. One consequence of the mathematical rules of quantum mechanics is a tradeoff in predictability between measurable quantities. The most famous form of this uncertainty principle says that no matter how a quantum particle is prepared or how carefully experiments upon it are arranged, it is impossible to have a precise prediction for a measurement of its position and also at the same time for a measurement of its momentum. Another consequence of the mathematical rules of quantum mechanics is the phenomenon of quantum interference, which is often illustrated with the double-slit experiment. In the basic version of this experiment, a coherent light source, such as a laser beam, illuminates a plate pierced by two parallel slits, and the light passing through the slits is observed on a screen behind the plate.

Chapter 2

The wave nature of light causes the light waves passing through the two slits to interfere, producing bright and dark bands on the screen – a result that would not be expected if light consisted of classical particles. However, the light is always found to be absorbed at the screen at discrete points, as individual particles rather than waves; the interference pattern appears via the varying density of these particle hits on the screen. Furthermore, versions of the experiment that include detectors at the slits find that each detected photon passes through one slit (as would a classical particle), and not through

both slits (as would a wave). However, such experiments demonstrate that particles do not form the interference pattern if one detects which slit they pass through. This behavior is known as wave-particle duality.

In addition to light, electrons, atoms, and molecules are all found to exhibit the same dual behavior when fired towards a double slit. Another non-classical phenomenon predicted by quantum mechanics is quantum tunnelling: a particle that goes up against a potential barrier can cross it, even if its kinetic energy is smaller than the maximum of the potential. In classical mechanics this particle would be trapped. Quantum tunnelling has several important consequences, enabling radioactive decay, nuclear fusion in stars, and applications such as scanning tunnelling microscopy, tunnel diode and tunnel field-effect transistor. When quantum systems interact, the result can be the creation of quantum entanglement: their properties become so intertwined that a description of the whole solely in terms of the individual parts is no longer possible. Erwin Schrödinger called entanglement "... the characteristic trait of quantum mechanics, the one that enforces its entire departure from classical lines of thought".

Quantum entanglement enables quantum computing and is part of quantum communication protocols, such as quantum key distribution and superdense coding. Contrary to popular misconception, entanglement does not allow sending signals faster than light, as demonstrated by the no-communication theorem. Another possibility opened by entanglement is testing for "hidden variables", hypothetical properties more fundamental than the quantities addressed in quantum theory itself, knowledge of which would allow more exact predictions than quantum theory provides. A collection of results, most significantly Bell's theorem, have demonstrated that broad classes of such hidden-variable theories are in fact incompatible with quantum physics. According to Bell's theorem, if nature actually operates in accord with any theory of local hidden variables, then the results of a Bell test will be constrained in a particular, quantifiable way.

Many Bell tests have been performed and they have shown results incompatible with the constraints imposed by local hidden variables. It is not possible to present these concepts in more than a superficial way without introducing the mathematics involved; understanding quantum mechanics requires not only manipulating complex numbers, but also linear algebra, differential equations, group theory, and other more advanced subjects. Accordingly, this article will present a mathematical formulation of quantum mechanics and survey its application to some useful and oft-studied examples.

== Mathematical formulation == In the mathematically rigorous formulation of quantum mechanics, the state of a quantum mechanical system is a vector ψ belonging to a (separable) complex Hilbert space

$\mathbf{H}$. This vector is postulated to be normalized under the Hilbert space inner product, that is, it obeys $\langle \psi, \psi \rangle = 1$, and it is well-defined up to a complex number of modulus 1 (the global phase), that is, $e^{i\alpha} \psi$ and $e^{i\alpha} \psi$ represent the same physical system. In other words, the possible states are points in the projective space of a Hilbert space, usually called the complex projective space.

Chapter 3

The exact nature of this Hilbert space is dependent on the system – for example, for describing position and momentum the Hilbert space is the space of complex square-integrable functions $L^2(\mathbb{R})$,

while the Hilbert space for the spin of a single proton is simply the space of two-dimensional complex vectors $\mathbb{C}^2$ with the usual inner product. Physical quantities of interest – position, momentum, energy, spin – are represented by observables, which are Hermitian (more precisely, self-adjoint) linear operators acting on the Hilbert space. A quantum state can be an eigenvector of an observable, in which case it is called an eigenstate, and the associated eigenvalue corresponds to the value of the observable in that eigenstate. More generally, a quantum state will be a linear combination of the eigenstates, known as a quantum superposition.

When an observable is measured, the result will be one of its eigenvalues with probability given by the Born rule: in the simplest case the eigenvalue λ is non-degenerate and the probability is given by $|\langle \psi, \vec{e}_\lambda \rangle|^2$, where $\vec{e}_\lambda$ is its associated unit-length eigenvector. More generally, the eigenvalue is degenerate and the probability is given by $\langle \psi, P_\lambda \psi \rangle$, where P_λ is the projector onto the eigenspace of λ .

$P_{\{\lambda\}}$ is the projector onto its associated eigenspace. In the continuous case, these formulas give instead the probability density.

After the measurement, if result λ was obtained, the quantum state is postulated to collapse to $|\lambda\rangle$. In the non-degenerate case, or to $P_{\lambda} \psi / \sqrt{\langle \psi | P_{\lambda} | \psi \rangle}$, in the general case. The probabilistic nature of quantum mechanics thus stems from the act of measurement. This is one of the most debated aspects of quantum theory, with different interpretations of quantum mechanics giving radically different answers to questions regarding quantum-state collapse, as discussed below.

Time evolution of a quantum state The time evolution of a quantum state is described by the Schrödinger equation:

$$i\hbar \frac{\partial}{\partial t} \psi = H \psi$$

Here H denotes the Hamiltonian, the observable corresponding to the total energy of the system, and $\hbar$ is the reduced Planck constant. The constant $\hbar$ is introduced so that the Hamiltonian is reduced to the classical Hamiltonian in cases where the quantum system can be approximated by a classical system; the ability to make such an approximation in certain limits is called the correspondence principle. The solution of this differential equation is given by $\psi(t) = e^{-iHt/\hbar} \psi(0)$.

The operator $U(t) = e^{-iHt/\hbar}$ is known as the time-evolution operator, and has the crucial property that it is unitary. This time evolution is deterministic in the sense that – given an initial quantum state $\psi(0)$ – it makes a definite prediction of what the quantum state $\psi(t)$ will be at any later time. Some wave functions produce probability distributions that are independent of time, such

as eigenstates of the Hamiltonian. Many systems that are treated dynamically in classical mechanics are described by such "static" wave functions.

Chapter 4

For example, a single electron in an unexcited atom is pictured classically as a particle moving in a circular trajectory around the atomic nucleus, whereas in quantum mechanics, it is described by a static wave function surrounding the nucleus. For example, the electron wave function for an unexcited hydrogen atom is a spherically symmetric function known as an s orbital (Fig. 1). Analytic solutions of the Schrödinger equation are known for very few relatively simple model Hamiltonians including the quantum harmonic oscillator, the particle in a box, the dihydrogen cation, and the hydrogen atom. Even the helium atom – which contains just two electrons – has defied all attempts at a fully analytic treatment, admitting no solution in closed form. However, there are techniques for finding approximate solutions. One method, called perturbation theory, uses the analytic result for a simple quantum mechanical model to create a result for a related but more complicated model by (for example) the addition of a weak potential energy.

Another approximation method applies to systems for which quantum mechanics produces only small deviations from classical behavior. These deviations can then be computed based on the classical motion. === Uncertainty principle === One consequence of the basic quantum formalism is the uncertainty principle. In its most familiar form, this states that no preparation of a quantum particle can imply simultaneously precise predictions both for a measurement of its position and for a measurement of its momentum. Both position and momentum are observables, meaning that they are

represented by Hermitian operators. The position operator $\hat{X}$ and momentum operator $\hat{P}$ do not commute, but rather satisfy the canonical commutation relation:

$$[\hat{X}, \hat{P}] = i\hbar$$

Given a quantum state, the Born rule lets us compute expectation values for both X and P , and moreover for powers of them. Defining the

uncertainty for an observable by a standard deviation, we have

$$\sigma_X = \sqrt{\langle X^2 \rangle - \langle X \rangle^2}$$

and likewise for the momentum:

$$\sigma_P = \sqrt{\langle P^2 \rangle - \langle P \rangle^2}.$$

The uncertainty principle states that

$$\sigma_X \sigma_P \geq \frac{\hbar}{2}.$$

Either standard deviation can in principle be made arbitrarily small, but not both simultaneously. This inequality generalizes to arbitrary pairs of self-adjoint operators A and B . The commutator of these two operators is

$$[A, B] = AB - BA$$

and this provides the lower bound on the product of standard deviations:

$$\sigma_A \sigma_B \geq \frac{1}{2} |\langle [A, B] \rangle|.$$

Another consequence of the canonical commutation relation is that the position and momentum operators are Fourier transforms of each other, so that a description of an object according to its momentum is the Fourier transform of its description according to its position.

The fact that dependence in momentum is the Fourier transform of the dependence in position means that the momentum operator is equivalent (up to an $i/\hbar$ factor) to taking the derivative according to the position, since in Fourier analysis differentiation corresponds to multiplication in the dual space. This is why in quantum

equations in position space, the momentum p_i is replaced by " $\frac{\partial}{\partial x_i}$ ", and in particular in the non-relativistic Schrödinger equation in position space the momentum-squared term is replaced with a Laplacian times $\frac{\hbar^2}{2m}$. Composite systems and entanglement == When two different quantum systems are considered together, the Hilbert space of the combined system is the tensor product of the Hilbert spaces of the two components.

Chapter 5

For example, let A and B be two quantum systems, with Hilbert spaces $\mathcal{H}_A$ and $\mathcal{H}_B$, respectively. The Hilbert space of the composite system is then $\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$.

If the state for the first system is the vector ψ_A and the state for the second system is ψ_B , then the state of the composite system is $\psi_A \otimes \psi_B$. Not all states in the joint Hilbert space $\mathcal{H}_{AB}$ can be written in this form, however, because the superposition principle implies that linear combinations of these "separable" or "product states" are also valid.

For example, if ψ_A and ϕ_A

ψ_A are both possible states for system A, and likewise ψ_B are both possible states for system B, then

$$\frac{1}{\sqrt{2}}(\psi_A + \psi_B)$$
 is a valid joint state that is not separable. States that are not separable are called entangled. If the state for a composite system is entangled, it is impossible to describe either component system A or system B by a state vector. One can instead define reduced density matrices that describe the statistics that can be obtained by making measurements on either component system alone. This necessarily causes a loss of information, though: knowing the reduced density matrices of the individual systems is not enough to reconstruct the state of the composite system. Just as density matrices specify the state of a subsystem of a larger system, analogously, positive operator-valued measures (POVMs) describe the effect on a subsystem of a measurement performed on a larger system.

POVMs are extensively used in quantum information theory. As described above, entanglement is a key feature of models of measurement processes in which an apparatus becomes entangled with the system being measured. Systems interacting with the environment in which they reside generally become entangled with that environment, a phenomenon known as quantum decoherence. This can explain why, in practice, quantum effects are difficult to observe in systems larger than microscopic.

Equivalence between formulations There are many mathematically equivalent formulations of quantum mechanics. One of the oldest and most common is the "transformation theory" proposed by Paul Dirac, which unifies and generalizes the two earliest formulations of quantum mechanics – matrix mechanics (invented by Werner Heisenberg) and wave mechanics (invented by Erwin Schrödinger).

An alternative formulation of quantum mechanics is Feynman's path integral formulation, in which a quantum-mechanical amplitude is considered as a sum over all possible classical and non-classical paths between the initial and final states. This is the quantum-mechanical counterpart of the action principle in classical mechanics. Symmetries and conservation laws The Hamiltonian H is known as the generator of time

evolution, since it defines a unitary time-evolution operator $U(t)$
$$U(t) = e^{-iHt/\hbar}$$
 for each value of t . From this relation between $U(t)$ and H , it follows that any observable A that commutes with H will be conserved: its expectation value will not change over time.

Chapter 6

This statement generalizes, as mathematically, any Hermitian operator A can generate a family of unitary operators parameterized by a variable t . Under the evolution generated by A , any observable B that commutes with A will be conserved. Moreover, if B is conserved by evolution under A , then A is conserved under the evolution generated by B .

This implies a quantum version of the result proven by Emmy Noether in classical (Lagrangian) mechanics: for every differentiable symmetry of a Hamiltonian, there exists a corresponding conservation law.

== Examples ==

Free particle

The simplest example of a quantum system with a position degree of freedom is a free particle in a single spatial dimension. A free particle is one which is not subject to external influences, so that its Hamiltonian consists only of its kinetic energy:

$$H = \frac{1}{2m} P^2$$

The general solution of the Schrödinger equation is given by

$$\psi(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{\psi}(k, 0) e^{i(kx - \frac{\hbar k^2}{2m}t)} dk,$$
 which is a superposition of all possible plane waves e^{ikx} . The coefficients of the superposition are $\hat{\psi}(k, 0)$, which is the Fourier transform of the initial quantum state $\psi(x, 0)$. It is not possible for the solution to be a single momentum eigenstate, or a single position eigenstate, as these are not normalizable quantum states.

Instead, we can consider a Gaussian wave packet:

$$\psi(x, 0) = \frac{1}{\sqrt{4\pi a}} e^{-\frac{x^2}{2a}},$$
 which has Fourier transform, and therefore momentum distribution

$$\hat{\psi}(k, 0) = \sqrt{a} e^{-\frac{ak^2}{2}}.$$
 We see that as we make a smaller the spread in position gets smaller, but the spread in momentum gets larger. Conversely, by

making a larger we make the spread in momentum smaller, but the spread in position gets larger. This illustrates the uncertainty principle. As we let the Gaussian wave packet evolve in time, we see that its center moves through space at a constant velocity (like a classical particle with no forces acting on it).

However, the wave packet will also spread out as time progresses, which means that the position becomes more and more uncertain. The uncertainty in momentum, however, stays constant.

Particle in a box The particle in a one-dimensional potential energy box is the most mathematically simple example where restraints lead to the quantization of energy levels. The box is defined as having zero potential energy everywhere inside a certain region, and therefore infinite potential energy everywhere outside that region. For the one-dimensional case in the x direction, the time-independent Schrödinger equation may be written

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2} = E \psi,$$

Chapter 7

With the differential operator defined by $\hat{p} = -i\hbar \frac{d}{dx}$

the previous equation is evocative of the classic kinetic energy analogue,

$$\frac{1}{2m} p^2 = E,$$

with state ψ in this case having energy E coincident with the kinetic energy of the particle. The general solutions of the Schrödinger equation for the particle in a box are

$$\psi_k(x) = A \sin(kx) + B e^{ikx}$$

$$\psi(x) = Ae^{ikx} + Be^{-ikx} \quad \text{or, from Euler's formula,} \quad \psi(x) = C \sin(kx) + D \cos(kx).$$

The infinite potential walls of the box determine the values of C , D , and k at $x=0$ and $x=L$ where ψ must be zero. Thus, at $x=0$,

$$\psi(0) = C \sin(0) + D \cos(0) = D = 0 \quad \text{and} \quad D = 0.$$

At $x=L$,

$$\psi(L) = C \sin(kL) = 0 \quad \text{in which } C \text{ cannot be zero as this would conflict with the postulate that } \psi \text{ has norm 1.}$$

Therefore, since $\sin(kL) = 0$, kL must be an integer multiple of π ,

$$kL = n\pi \quad \text{where } n = 1, 2, 3, \dots$$

This constraint on k implies a constraint on the energy levels, yielding

$$E_n = \frac{\hbar^2 k^2}{2m} = \frac{\hbar^2 n^2 \pi^2}{2mL^2}$$

$$E_n = \frac{n^2 \pi^2 \hbar^2}{8mL^2}.$$

A finite potential well is the generalization of the infinite potential well problem to potential wells having finite depth. The finite potential well problem is mathematically more complicated than the infinite particle-in-a-box problem as the wave function is not pinned to zero at the walls of the well. Instead, the wave function must satisfy more complicated mathematical boundary conditions as it is nonzero in regions outside the well.

Another related problem is that of the rectangular potential barrier, which furnishes a model for the quantum tunneling effect that plays an important role in the performance of modern technologies such as flash memory and scanning tunneling microscopy. === Harmonic oscillator === As in the classical case, the potential for the quantum harmonic oscillator is given by $V(x) = \frac{1}{2} m \omega^2 x^2$. This problem can either be treated by directly solving the Schrödinger equation, which is not trivial, or by using the more elegant "ladder method" first proposed by Paul Dirac.

Chapter 8

The eigenstates are given by $\psi_n(x) = \frac{1}{\sqrt{2^n n!}} \left(\frac{m\omega}{\pi \hbar} \right)^{1/4} e^{-\frac{m\omega x^2}{2\hbar}} H_n \left(\sqrt{\frac{m\omega}{\hbar}} x \right)$.

$$\psi_n(x) = \sqrt{\frac{1}{2^n n!}} \left(\frac{m\omega}{\pi \hbar} \right)^{1/4} e^{-\frac{m\omega x^2}{2\hbar}} \cdot H_n \left(\sqrt{\frac{m\omega}{\hbar}} x \right),$$

$$\psi_n = 0, 1, 2, \dots$$

where H_n are the Hermite polynomials H_n

$$\left(\frac{x}{\sigma} \right)^n = \left(\frac{x}{\sigma} \right)^n$$

$$e^{-\frac{x^2}{2\sigma^2}} \frac{d^n}{dx^n} e^{-\frac{x^2}{2\sigma^2}}$$

$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}$$

and the corresponding energy levels are

$$E_n = \hbar \omega \left(n + \frac{1}{2} \right)$$

This is another example illustrating the discretization of energy for bound states. == Mach–Zehnder interferometer == The Mach–Zehnder interferometer (MZI) illustrates the concepts of superposition and interference with linear algebra in dimension 2, rather than differential equations. It can be seen as a simplified version of the double-slit experiment, but it is of interest in its own right, for example in the delayed choice quantum eraser, the Elitzur–Vaidman bomb tester, and in studies of quantum entanglement.

We can model a photon going through the interferometer by considering that at each point it can be in a superposition of only two paths: the "lower" path which starts from the left, goes straight through both beam splitters, and ends at the top, and the "upper" path which starts from the bottom, goes straight through both beam splitters, and ends at the right. The quantum state of the photon is therefore a vector $\psi \in \mathbb{C}^2$ that is a superposition of the "lower" path and the "upper" path.

$$\psi = \begin{pmatrix} \psi_0 \\ \psi_1 \end{pmatrix}$$

$$U = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$
 and the "upper" path ψ_u is, $\langle \psi_u | \psi_u \rangle = 1$; $\langle \psi_l | \psi_l \rangle = 1$ for complex α, β ; $\langle \psi | \psi \rangle = 1$. In order to respect the postulate that $\langle \psi | \psi \rangle = 1$ we require that $|\alpha|^2 + |\beta|^2 = 1$.

Both beam splitters are modelled as the unitary matrix $B = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$, which means that when a photon meets the beam splitter it will either stay on the same path with a probability amplitude of $1/\sqrt{2}$, or be reflected to the other path with a probability amplitude of $i/\sqrt{2}$. The phase shifter on the upper arm is modelled as the unitary matrix $P = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\Delta\Phi} \end{pmatrix}$, which means that if the photon is on the "upper" path it will gain a relative phase of $\Delta\Phi$; and it will stay unchanged if it is in the lower path. A photon that enters the interferometer from the left will then be acted upon with a beam splitter B , a phase shifter P , and another beam splitter B , and so end up in the state $B P B | \psi \rangle = | \psi \rangle$.

$$p(l) = |\langle \psi_1, \text{BPB} \psi_1 \rangle|^2 = \sin^2 \left\{ \frac{\Delta \Phi}{2} \right\}.$$

One can therefore use the Mach–Zehnder interferometer to estimate the phase shift by

$\{2\}$. } One can therefore use the Mach–Zehnder interferometer to estimate the phase shift by estimating these probabilities. It is interesting to consider what would happen if the photon were definitely in either the "lower" or "upper" paths between the beam splitters. This can be accomplished by blocking one of the paths, or equivalently by removing the first beam splitter (and feeding the photon from the left or the bottom, as desired). In both cases, there will be no interference between the paths anymore, and the probabilities are given by
$$p(u) = p(l) = 1/2$$
 , independently of the phase Φ .

Chapter 9

From this we can conclude that the photon does not take one path or another after the first beam

splitter, but rather that it is in a genuine quantum superposition of the two paths. == Applications == Quantum mechanics has had enormous success in explaining many of the features of our universe, with regard to small-scale and discrete quantities and interactions which cannot be explained by classical methods. Quantum mechanics is often the only theory that can reveal the individual behaviors of the subatomic particles that make up all forms of matter (electrons, protons, neutrons, photons, and others). Solid-state physics and materials science are dependent upon quantum mechanics. In many aspects, modern technology operates at a scale where quantum effects are significant.

Important applications of quantum theory include quantum chemistry, quantum optics, quantum computing, superconducting magnets, light-emitting diodes, the optical amplifier and the laser, the transistor and semiconductors such as the microprocessor, medical and research imaging such as magnetic resonance imaging and electron microscopy. Explanations for many biological and physical phenomena are rooted in the nature of the chemical bond, most notably the macro-molecule DNA. == Relation to other scientific theories == === Classical mechanics === The rules of quantum mechanics assert that the state space of a system is a Hilbert space and that observables of the system are Hermitian operators acting on vectors in that space – although they do not tell us which Hilbert space or which operators.

These can be chosen appropriately in order to obtain a quantitative description of a quantum system, a necessary step in making physical predictions. An important guide for making these choices is the correspondence principle, a heuristic which states that the predictions of quantum mechanics reduce to those of classical mechanics in the regime of large quantum numbers. One can also start from an established classical model of a particular system, and then try to guess the underlying quantum model that would give rise to the classical model in the correspondence limit. This approach is known as quantization. When quantum mechanics was originally formulated, it was applied to models whose correspondence limit was non-relativistic classical mechanics. For instance, the well-known model of the quantum harmonic oscillator uses an explicitly non-relativistic expression for the kinetic energy of the oscillator, and is thus a quantum version of the classical harmonic oscillator.

Complications arise with chaotic systems, which do not have good quantum numbers, and quantum chaos studies the relationship between classical and quantum descriptions in these systems. Quantum decoherence is a mechanism through which quantum systems lose coherence, and thus become incapable of displaying many typically quantum effects: quantum superpositions become simply probabilistic mixtures, and quantum entanglement becomes simply classical correlations. Quantum

coherence is not typically evident at macroscopic scales, though at temperatures approaching absolute zero quantum behavior may manifest macroscopically. Many macroscopic properties of a classical system are a direct consequence of the quantum behavior of its parts.

For example, the stability of bulk matter (consisting of atoms and molecules which would quickly collapse under electric forces alone), the rigidity of solids, and the mechanical, thermal, chemical, optical and magnetic properties of matter are all results of the interaction of electric charges under the rules of quantum mechanics. === Special relativity and electrodynamics === Early attempts to merge quantum mechanics with special relativity involved the replacement of the Schrödinger equation with a covariant equation such as the Klein–Gordon equation or the Dirac equation. While these theories were successful in explaining many experimental results, they had certain unsatisfactory qualities stemming from their neglect of the relativistic creation and annihilation of particles. A fully relativistic quantum theory required the development of quantum field theory, which applies quantization to a field (rather than a fixed set of particles).

Chapter 10

The first complete quantum field theory, quantum electrodynamics, provides a fully quantum description of the electromagnetic interaction. Quantum electrodynamics is, along with general relativity, one of the most accurate physical theories ever devised. The full apparatus of quantum field theory is often unnecessary for describing electrodynamic systems. A simpler approach, one that has been used since the inception of quantum mechanics, is to treat charged particles as quantum mechanical objects being acted on by a classical electromagnetic field. For example, the elementary quantum model of the hydrogen atom describes the electric field of the hydrogen atom using a classical

$$V(r) = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2} \quad \text{Coulomb potential.}$$

Likewise, in a Stern–Gerlach experiment, a charged particle is modeled as a quantum system, while the background magnetic field is described classically. This "semi-classical" approach fails if quantum fluctuations in the electromagnetic field play an important role, such as in the emission of photons by charged particles. Quantum field theories for the strong nuclear force and the weak nuclear force have

also been developed. The quantum field theory of the strong nuclear force is called quantum chromodynamics, and describes the interactions of subnuclear particles such as quarks and gluons. The weak nuclear force and the electromagnetic force were unified, in their quantized forms, into a single quantum field theory (known as electroweak theory), by the physicists Abdus Salam, Sheldon Glashow and Steven Weinberg.

=== Relation to general relativity === Even though the predictions of both quantum theory and general relativity have been supported by rigorous and repeated empirical evidence, their abstract formalisms contradict each other and they have proven extremely difficult to incorporate into one consistent, cohesive model. Gravity is negligible in many areas of particle physics, so that unification between general relativity and quantum mechanics is not an urgent issue in those particular applications. However, the lack of a correct theory of quantum gravity is an important issue in physical cosmology and the search by physicists for an elegant "Theory of Everything" (TOE). Consequently, resolving the inconsistencies between both theories has been a major goal of 20th- and 21st-century physics. This TOE would combine not only the models of subatomic physics but also derive the four fundamental forces of nature from a single force or phenomenon.

One proposal for doing so is string theory, which posits that the point-like particles of particle physics are replaced by one-dimensional objects called strings. String theory describes how these strings propagate through space and interact with each other. On distance scales larger than the string scale, a string looks just like an ordinary particle, with its mass, charge, and other properties determined by the vibrational state of the string. In string theory, one of the many vibrational states of the string corresponds to the graviton, a quantum mechanical particle that carries gravitational force. Another popular theory is loop quantum gravity (LQG), which describes quantum properties of gravity and is thus a theory of quantum spacetime. LQG is an attempt to merge and adapt standard quantum mechanics and standard general relativity. This theory describes space as an extremely fine fabric "woven" of finite loops called spin networks.

The evolution of a spin network over time is called a spin foam. The characteristic length scale of a spin foam is the Planck length, approximately 1.616×10^{-35} m, and so lengths shorter than the Planck length are not physically meaningful in LQG. == Philosophical implications == Since its inception, the many counter-intuitive aspects and results of quantum mechanics have provoked strong philosophical debates and many interpretations. The arguments centre on the probabilistic nature of quantum mechanics, the difficulties with wavefunction collapse and the related measurement problem, and quantum nonlocality. Perhaps the only consensus that exists about these issues is that there is no

consensus. Richard Feynman once said, "I think I can safely say that nobody understands quantum mechanics. " According to Steven Weinberg, "There is now in my opinion no entirely satisfactory interpretation of quantum mechanics.

Chapter 11

" The views of Niels Bohr, Werner Heisenberg and other physicists are often grouped together as the "Copenhagen interpretation". According to these views, the probabilistic nature of quantum mechanics is not a temporary feature which will eventually be replaced by a deterministic theory, but is instead a final renunciation of the classical idea of "causality".

